

Algebra II with Trigonometry

Chapter 5.3

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:

[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Problems

1. If $\sin \theta + \cos \theta = \frac{7}{5}$, determine without a calculator which quadrant(s) θ could lie in.
2. Find all values of k such that the equation $\tan(k\pi + \frac{\pi}{6}) = \tan(\frac{5\pi}{6} - k\pi)$ holds for integer k between -10 and 10 .
3. Without explicit calculation, compare the areas of two triangles: Triangle A with sides 10 and 24 and included angle 60° , and Triangle B with sides 12 and 20 and included angle 90° . Which triangle has a larger area?
4. Let θ be an angle in standard position with $\cos \theta = -\frac{4}{5}$ and θ in the third quadrant. Compute the exact value of $\tan 2\theta$.
5. Let α satisfy $\tan \alpha = 2$ and α is in the second quadrant. What is the value of $\sin(2\alpha)$?
6. An observer is 1 mile from a launch pad. Two rockets are launched at the same time. Rocket A is directly overhead at an angle of elevation that increases from 30° to 60° in 10 seconds. Rocket B's angle of elevation increases from 45° to 75° in the same time. Which rocket had the greater average rate of increase in vertical height, and by how much? (Give an exact or simplified answer.)